Piyush Kumar



BTech AI & ML, ADGIPS (2022–26) · Backend, mathematical modelling, technical writing · Stepping into mindfulness teaching
Live work: bramble.co.in · GitHub: pkapka-org
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> *"A plane can fly on autopilot, but passengers still rest easier with a skilled pilot in the seat."*

PROFILE

Student and practitioner in backend systems, mathematical modelling and clear technical writing. I design and build software with careful use of AI coding assistants and write mathematical notes in LaTeX. I also practise mindfulness daily and am now starting to guide beginners in it.

EXPERIENCE

Blueberry Travel Solutions Pvt Ltd · Backend Supervisor and Visiting Full-Stack Manager (current)

- Previously Emergency Full-Stack Engineer.
- Backend design and live production work at bramble.co.in.

INTERNSHIP

India Meteorological Department · Satellite Division

Mausam Bhavan, Lodi Road, New Delhi · ended 31 December 2025 · Supervisors: Dr R.K. Giri and Mr Yogesh Kumar Jha

- Worked on visualisation software for satellite data in webcast mode.

PROJECTS

Secure online CBT system (Django), completed July 2026

- AES-256-GCM encryption, 2-of-3 Shamir key unlock, signed receipts, 32 passing tests. Code kept private for security.

Mathematical model of a filter-based air purifier

- Closed-room model treating pollutants as a solute, with step-by-step concentration updates from filter volume, capture and clogging. LaTeX note with simulation.

Traffic model

- Mathematical model of traffic.

Randomness framework

- Defines randomness as deviation from predictability against a fixed reference using "stencils". LaTeX note.

Iterative square-root method

- Independently derived an iterative square-root method that matches Heron's method.

MINDFULNESS (STEPPING INTO TEACHING)

I practise a simple way of being mindful that I call Universe Watching. You notice the world around you as it happens, sometimes relaxed and sometimes focused, and let it settle into quiet meditation. I've done it daily on my mindfulness runs since 2024 and am now offering beginner sessions online over Zoom, Google Meet or WhatsApp.

1. **Watch:** notice sights, sounds and feelings without judging them.
2. **Balance:** switch between relaxed noticing and gentle focus.
3. **Settle:** let it turn into a few minutes of still meditation.

SKILLS

- **Programming:** Python (including OOP); can read C++.
- **Technical:** machine learning, NLP, algorithm design, mathematical modelling.
- **AI workflow:** combining AI tools in one workflow; runs a three-agent AI team for freelance delivery.
- **Freelance services:** copywriting, Excel data cleaning, statistics, LaTeX typesetting, laser and Cricut files, 3D printable design, FEM design optimisation, backend and ML consulting.

EDUCATION

BTech in Artificial Intelligence and Machine Learning, ADGIPS

2022–2026 · CGPA about 7

BOOK

The Sapling: There Is Always a Way, a novel · Amazon